Math 5050 – Special Topics: Algebraic Topology– Spring 2026

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Notes and Observations

Notes and Observations

LOOK UP ROBERT GHRIST PAPERS, EULER CHARACTERISTIC BETTI NUMBER HOW THESE CONTRIBUTE TO HOMOLOGY

Helpful Diagrams

Topological Objects

- n-Ball: unit ball in $\mathbb{R}^n$.
  Line (1-ball): $D^1 = \{x : x^2 \leq 1\}$
  Disk (2-ball): $D^2 = \{(x, y) : x^2 + y^2 \leq 1\}$
  n-ball: $D^n = \{(x_1, \ldots, x_n) : \sum x_i^2 \leq 1\}$
  'e-cell': interior of n-ball, $e^n = \{(x_1, \ldots, x_n) : \sum x_i^2 < 1\}$
- $n-1$-surface: $S^{n-1} = \partial D^n = \{(x_1, \ldots, x_n) : \sum x_i^2 = 1\}$
  Two-point: $S^0 = \partial D^1 = \{0, 1\}$.
  Cirlce: $S^1 = \partial D^2 = \{(x, y) : x^2 + y^2 = 1\}$.
  Sphere: $S^2 = \partial D^3 = \{(x, y, z) : x^2 + y^2 + z^2 = 1\}$.
- A ***cone*** on X is the quotient space $cX = (X \times I)/(X \times \{1\})$.
  Every element in cX is (X, t) where $0 \leq t \leq 1$ and $(X, 1)$ is the identity element.

Invariant Topological Properties

- Connectedness.
- Contractibility.
- Compactness.
- Completeness.

Operators on topological objects

1. **Product** $X \times Y$. For every point in X we 'attach' a copy of Y.
   Boundary of Product: $\partial(X \times Y) = (\partial X \times Y) \cup (X \times \partial Y)$
   Beware $S^1 \times S^1$ can describe sphere as well as a torus but they aren't isomorphic.
2. **Quotient** Y/X
   Take X to be a point and map all of Y to it.
3. **Suspensions**
4. **Join**

5. **Wedge Sum**
6. **Smash Product**

Vocabulary from Topology

- **Properly Discontinuous**
  A properly discontinuous action of a group G on a topological space X is a group action where for every point $x \in X$, there exists a neighborhood U such that the set of group elements $g \in G$ moving U to intersect itself is limited to the identity, i.e., $gU \cap U \neq \emptyset$ only for $g = e$. This ensures the quotient map $X \to X/G$ is a covering map.
- **Connected**
  A topological space X is said to be *disconnected* if it is the union of two disjoint non-empty open sets. Otherwise, X is said to be ***connected.***
- **Simply Connected**
  A topological space is simply connected if it is path-connected and every continuous loop within it can be continuously shrunk to a point (i.e., trivial group).
- **Path Connected**
  A path from a point x to a point y in a topological space X is a continuous function f from the unit interval $[0, 1]$ to X with $f(0) =$ and $f(1) = y$. A path-component of X is an equivalence class of X under the equivalence relation which makes x equivalent to y if and only if there is a path from x to y. The space X is said to be path-connected (or pathwise connected or **0**-connected) if there is exactly one path-component.
- **Locally Path Connected**
  A space X is called locally connected at x if every neighborhood of x contains a connected open neighborhood of x, that is, if the point x has a neighborhood base consisting of connected open sets.
- **Semi-locally Path Connected**
  A space X is called semi-locally simply connected if every point x in X and every neighborhood V of x has an open neighborhood U of x such that $x \in U \subset V$ with the property that every loop in U can be contracted to a single point within X (i.e. every loop in U is nullhomotopic in X).
- **Homeomorphic**
  Give two topological spaces X and Y. They are said to be ***homeomorphic*** if there exists a continuous function $f : X \to Y$ and its inverse. (think "isomorphsim" but in topological terms).
- **Locally Homeomorphic**
- **Quotient Topology**

Deformation Retract

Definition 0.0.1 (Deformation Retract)**.** A subspace A of X is ***deformation retract*** of X if there is a retraction $g : X \to A$ (i.e., a map $g : X \to A$ with $g(a) = a$ for every $a \in A$) such that g is homotopic to the identiy map $id : X \to X$. A subspace A of X is a ***strong deformation retract*** if there is a retraction $g : X \to A$ such that g is homotopic rel A to the identity map $id : X \to X$.

Homotopy Equivalence and Deformation Retract

Homotopy: Two maps $f_0 : (X, A) \to (Y, B)$ and $f_1 : (X, A) \to (Y, B)$ are ***homotopic***, i.e., $f_0 \sim f_1$, if there is a map

$$F : (X, A) \times I \to (Y, B)$$

where $F(x, 0) = f_0$ and $F(x, 1) = f_1$ for every $x \in X$. *Homotopy* is an equivalence relation.

Definition 0.0.2 (Homotopy Equivalence)**.** A continuous map $f : X \to Y$ is a ***homotopy equivalence*** if there exists a continuous map $g : Y \to X$ such that $f \circ g = \mathbb{I}_Y$ and $g \circ f = \mathbb{I}_X$. This means the maps are inverses of each other up to homotopy, implying that the spaces X and Y share the same homotopy type.

Theorem 0.0.3. *Put simply two spaces are homotopy equivalent if there is a third space that deformation retracts (shrinks/expands) to both.*

Lemma 0.0.4. *A cell complex, X and a subcomplex A, then the quotient map $X \to X/A$ is a homotopy equivalence.*

Theorem 0.0.5 (Homeomorphism implies Isomorphism (not visa versa))**.** *If X is homeomorphic to Y then $\pi_1(X) \cong \pi_1(Y)$.*

Theorem 0.0.6 (Homotopy Equivalence implies Isomorphism)**.**

Remark 0.0.7. Understand a progression of relationships between two topological spaces X, Y from"isomorphic/homeomorphic" $\to$ "deformation retract between X and Y" $\to$ "homotopy equivalent".

Further: X and Y need only be a deformation retract between them.

The Fundamental Group

The ***Fundamental Group*** is the set of homotopy classes of loops.

Theorem 0.0.8 (The Fundamental Group is a group)**.** .

The Fundamental Group $\pi_1(X, x_0)$ is a group

Theorem 0.0.9. .

$\pi_1(S^1) \cong \mathbb{Z}$[1]

Proof: Let a be the generator of $\pi_1(S^1)$, i.e., $\pi_1(S^1) = \langle a \rangle$. Define $\varphi : \mathbb{Z} \to \pi_1(S^1)$ as

$$\varphi(n) = a^n$$

Then

- *φ is surjective, i.e., $\varphi(m) = \varphi(n) \implies m = n$.*
- *φ is injective, $\forall [f] \in S^1 \, \exists n \in \mathbb{Z} \to \varphi(n) = a^n = f$*

Thus, an isomorphism exists between $\mathbb{Z}$ and $\pi_1(S^1)$

Proof (outline): Show that there exists a homomorphism $\varphi : \mathbb{Z} \to \pi_1(S^1)$ then show that

- *φ is surjective, i.e., $\varphi(m) = \varphi(n) \implies m = n$.*
- *φ is injective, $\forall [f] \in S^1 \, \exists n \in \mathbb{Z} \to \varphi(n) = f$*

Remark 0.0.10. In the video *Algebraic Topolgy 3: The Fundamental Group is a Group*, the instructor uses a helix to describe multiple but separate paths around S^1 indicating that a full loop in S^1 can be a path on the helix from a point to the same point on the next (or successive) level. The helix is eventually mapped to $\mathbb{R}$ where the closed loops are then mapped to $\mathbb{Z}$.

Covering Space

If $\tilde{X}$ is a covering space for X then $\pi_1(\tilde{X})$ is a subgroup of $\pi_1(X)$. And there is a bijection of the set/space of maps $\tilde{X} \to X$ and subgroups of $\pi_1(X)$.

A symmetric foundamental group is normal

[1] Note that this is independent of base-point x_0.

Definition 0.0.11 (Covering Space)**.**

$$\begin{aligned} p : \tilde{X} &\to x \\ \tilde{x_0} &\mapsto x_0 \text{ basepoint } (X, x_0) \end{aligned}$$

open covers $\{u_\alpha\}$ of X such that $p^{-1}(\alpha)$ is homeomorphic to u_α.

There is an induced mapping $p_* : \pi_1(\tilde{X}, \tilde{X}_0) \to \pi_1(X, x_0)$ where p_* is a subgroup of $\pi_1(X, x_0)$.

Remark 0.0.12. It is the projection, p, along with the preimage homeomorphism that forms this connection between Covering Spaces and Fundamental Groups.

Simplicial Diagrams

Torus

$$\begin{tikzcd} v \arrow[r, "b"] & v \\ v \arrow[u, "a"] \arrow[ur, "c"] \arrow[r, "b"'] & v \arrow[u, "a"'] \end{tikzcd}$$

Homology

Simplicial Mapping, Δ^n

The ***simplicial mapping***, $\Delta_n(X)$, is the set of all mappings of n-simplexes onto the topological space X. These mappings form a group.

Chain Complex

Given a set of nested groups C_n such that

$$\cdots \to C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots \longrightarrow C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} \to 0$$

Where the image of $\operatorname{img} \partial_{n+1} \subset \ker \partial_n$ for all n. We define the ***exact chain*** if $\operatorname{img} \partial_{n+1} = \ker \partial_n$ for all n.

The nth simplicial homology group, $H_n^{\Delta}(X)$

The chain complex where $C_n = \Delta_n$ of simplicial elements. Where each element is defined as $H_n(X) = \ker \partial_n / \operatorname{img} \partial_{n+1}$.

The nth singular homology group, $H_n(X)$

The chain complex where C_n is the set of continuous functions on X in n variables. Where each element is defined as $H_n(X) = \ker \partial_n / \operatorname{img} \partial_{n+1}$.

Exact Chain Complexes

Small Exact Chains

Define $i : A \to B$ and $j : B \to C$ and

$$0 \longrightarrow A \xrightarrow{\quad i \quad} B \xrightarrow{\quad j \quad} C \longrightarrow 0$$

$$\ker(i) = 0 \longrightarrow \operatorname{img}(i) = \ker(j) \longrightarrow \operatorname{img}(j) = C$$

where i is injective and j is surjective

$$C \cong B/\ker(j) = B/\operatorname{Im}(i) \cong B/A$$

Example:

$$0 \longrightarrow 2\mathbb{Z} \xrightarrow{i} \mathbb{Z} \xrightarrow{j} \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

i is the 'inclusion map', e.g., $i(4) = 4$ (i.e., injective), j is modulus, i.e., mapping odd numbers to 1 and even 0, surjective. Notice that the image of i are even numbers which are the kernel of j.

Definition 0.0.13. Define $\hat{H}_n(X)$ as $H_n(X)$ excluding $n = 0$.

Theorem 0.0.14. *Define the short chain $\mathcal{C}_n$ as*

$$\ldots \xrightarrow{\partial} \hat{H}_n(A) \xrightarrow{i_*} \hat{H}_n(X) \xrightarrow{q_*} \hat{H}_n(X/A) \xrightarrow{\partial} \ldots$$

where i_* is the induced map from $i : A \hookrightarrow X$, the inclusion map, and q_* is the induced map from the quotient map $q : X \to X/A$ and ∂ will be boundary maps (TBD). Notice that

$$\ldots \xrightarrow{\partial} \mathcal{C}_{n+1} \xrightarrow{\partial} \mathcal{C}_n \xrightarrow{\partial} \mathcal{C}_{n-1} \xrightarrow{\partial} \ldots$$

Short Exact Sequence of Chains

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\ldots \arrow[r, "\partial"] & \mathcal{A}_{n+1} \arrow[r, "\partial"] \arrow[d, "i"] & \mathcal{A}_n \arrow[r, "\partial"] \arrow[d, "i"] & \mathcal{A}_{n-1} \arrow[r, "\partial"] \arrow[d, "i"] & \ldots \\
\ldots \arrow[r, "\partial"] & \mathcal{B}_{n+1} \arrow[r, "\partial"] \arrow[d, "j"] & \mathcal{B}_n \arrow[r, "\partial"] \arrow[d, "j"] & \mathcal{B}_{n-1} \arrow[r, "\partial"] \arrow[d, "j"] & \ldots \\
\ldots \arrow[r, "\partial"] & \mathcal{C}_{n+1} \arrow[r, "\partial"] & \mathcal{C}_n \arrow[r, "\partial"] & \mathcal{C}_{n-1} \arrow[r, "\partial"] & \ldots \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

Regarding homologies

Define the following

$$\begin{aligned}
Z_n(X) &= \ker(\partial_n : C_n(X) \to C_{n-1}(X)), \text{ the group of } \textit{singular } n\textit{-cycles} \\
B_n(X) &= \operatorname{Im}(\partial_{n+1} : C_{n+1}(X) \to C_n(X)), \text{ the group of } \textit{singular } n\textit{-boundaries} \\
H_n(X) &= Z_n(X)/B_n(X), \text{ the } n^{\text{th}} \textit{ singular homology group} \text{ of } X
\end{aligned}$$

Explain $H_n(X)$

The Chain Group $C_n(X)$ To understand $H_n(X)$, we start with the singular n-chain group, $C_n(X)$. This is the free abelian group generated by all singular n-simplices in the space X. An element of this group is a formal finite sum $\sum_i n_i \sigma_i$, where each σ_i is a continuous map from the standard n-simplex Δ^n to X.

The Boundary Operator ∂_n The boundary operator $\partial_n : C_n(X) \to C_{n-1}(X)$ is a homomorphism that takes an n-simplex and maps it to an alternating sum of its $(n-1)$-dimensional faces. A fundamental property of this operator is that $\partial_n \circ \partial_{n+1} = 0$, which means that the boundary of a boundary is always zero.

Singular n-cycles $Z_n(X)$ The group of singular n-cycles, denoted by $Z_n(X)$, is the kernel of the boundary operator:

$$Z_n(X) = \ker(\partial_n) = \{c \in C_n(X) \mid \partial_n(c) = 0\}$$

These represent chains that have no boundary (they are "closed").

Singular n-boundaries $B_n(X)$ The group of singular n-boundaries, denoted by $B_n(X)$, is the image of the boundary operator from the dimension above:

$$B_n(X) = \mathrm{im}(\partial_{n+1}) = \{c \in C_n(X) \mid c = \partial_{n+1}(d) \text{ for some } d \in C_{n+1}(X)\}$$

Because $\partial_n \circ \partial_{n+1} = 0$, every boundary is necessarily a cycle ($B_n(X) \subseteq Z_n(X)$).

The Homology Group $H_n(X)$ The n-th singular homology group $H_n(X)$ is defined as the quotient group of cycles modulo boundaries:

$$H_n(X) = Z_n(X)/B_n(X)$$

Intuitively, $H_n(X)$ measures the presence of "n-dimensional holes" in the space. It counts cycles that are not boundaries of any $(n+1)$-dimensional shape. If two cycles differ by a boundary, they represent the same element (homology class) in $H_n(X)$.

AI Questions:

What are the properties of Covering Spaces?

A covering space is a concept deeply connected to the fundamental group. Let's explore its properties.

Let $p : E \to B$ be a continuous surjective map between topological spaces. We call p a ***covering map***, and E a ***covering space*** of B, if it satisfies the following property:

For every point $b \in B$, there exists an open neighborhood U of b in B such that its preimage $p^{-1}(U)$ is a disjoint union of open sets V_i in E, and for each i, the restriction of p to V_i, denoted $p|_{V_i} : V_i \to U$, is a homeomorphism. The neighborhood U is called an ***evenly covered*** neighborhood.

Key Properties of Covering Spaces:

1. **Local Homeomorphism:** A covering map $p : E \to B$ is a local homeomorphism. This means that for every point $e \in E$, there is a neighborhood of e that is mapped homeomorphically onto its image in B. This follows directly from the definition of an evenly covered neighborhood.
2. **Lifting Properties:** Covering spaces have unique lifting properties for paths and homotopies. This is where the connection to the fundamental group becomes crucial.
   - **Path Lifting Property:** Given a path $\gamma : [0, 1] \to B$ in the base space and a starting point $e_0 \in E$ such that $p(e_0) = \gamma(0)$, there exists a unique path $\tilde{\gamma} : [0, 1] \to E$ in the covering space that starts at e_0 (i.e., $\tilde{\gamma}(0) = e_0$) and covers γ (i.e., $p \circ \tilde{\gamma} = \gamma$). This lifted path $\tilde{\gamma}$ is called a ***lift*** of γ.
   - **Homotopy Lifting Property:** If two paths in B are homotopic, their lifts in E starting at the same point will also be homotopic (specifically, they will end at the same point).
3. **Relationship with the Fundamental Group:** The connection between covering spaces and the fundamental group is one of the most important results in algebraic topology.
   - Let $p : (E, e_0) \to (B, b_0)$ be a covering map. The induced homomorphism $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective. This means the fundamental group of the covering space is isomorphic to a subgroup of the fundamental group of the base space.
   - The set of lifts of a loop in B based at b_0 corresponds to the elements of the subgroup $p_*(\pi_1(E, e_0))$. Specifically, a loop γ in B lifts to a loop in E if and only if its homotopy class $[\gamma]$ is in the subgroup $p_*(\pi_1(E, e_0))$.
4. **Classification of Covering Spaces:** For a 'nice' space B (path-connected, locally path-connected, and semilocally simply-connected), there is a one-to-one correspondence between the subgroups of its fundamental group $\pi_1(B, b_0)$ and the isomorphism classes of based covering spaces of B. The trivial subgroup $\{e\}$ corresponds to a special covering space called the **universal covering space**, which is simply connected (i.e., its fundamental group is trivial).
5. **From another query: How is a Covering Space Related to a subgroup of a Fundamental Group?**. The terminology is a misnomer. The word 'covering' implies 'larger than' as in the definition of 'compact' has 'finite subcover' and whose union is larger. A covering space is much more related to a subgroup which IS NOT be as large as the original group.

Describe the homology of a torus.

To describe the homology of a torus, we consider the standard 2-torus, denoted as T^2 or $S^1 \times S^1$. The homology groups $H_n(T^2; \mathbb{Z})$ provide information about the "holes" of various dimensions within the shape.

$$\begin{tikzcd} v \arrow[r, "b"] & v \\ v \arrow[u, "a"] \arrow[r, "b"'] & v \arrow[u, "a"'] \end{tikzcd}$$

Step-by-Step Explanation

1. **Definition and CW Structure**
   A torus can be constructed by taking a square and identifying opposite edges. This gives it a CW complex structure consisting of:
   - **One 0-cell (vertex):** Let's call it v.
   - **Two 1-cells (edges):** Let's call them a and b. Both start and end at v.
   - **One 2-cell (face):** Let's call it f. The boundary of this face is attached along the path $aba^{-1}b^{-1}$ (the commutator of the two edges).
2. **Calculating $H_0(T^2)$ (Connectedness)**
   The 0th homology group represents the number of path-connected components. Since the torus is a single connected piece, the 0th homology group is the free abelian group of rank 1.
   Result: $H_0(T^2) \cong \mathbb{Z}$.
3. **Calculating $H_1(T^2)$ (1-Dimensional Holes)**
   The 1st homology group represents the "loops" that cannot be shrunk to a point or expressed as boundaries of a surface.
   - From the CW structure, the boundary of the 2-cell f is $\partial f = a + b - a - b = 0$ in the abelian chain group.
   - Since the boundary is zero, the two 1-cells a and b are cycles and are not boundaries of any 2-cell.
   - These two cycles correspond to the "meridional" and "longitudinal" circles of the torus.
   - **Result:** $H_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ (or $\mathbb{Z}^2$).
4. **Calculating $H_2(T^2)$ (2-Dimensional Voids)**
   The 2nd homology group represents enclosed volumes or "cavities."
   - The torus is a compact, connected, oriented 2-manifold without boundary.
   - For any such manifold of dimension n, the top homology group H_n is isomorphic to $\mathbb{Z}$.
   - In our CW complex, the single 2-cell f has a boundary of 0, making it a cycle. There are no 3-cells to serve as boundaries for it.

   * **Result:** $H_2(T^2) \cong \mathbb{Z}$.
5. **Higher Homology Groups**
   Since the torus is a 2-dimensional manifold, there are no cells of dimension 3 or higher.
   * **Result:** $H_n(T^2) = 0$ for all $n \geq 3$.

Final Answer Summary

The homology groups of the torus T^2 with integer coefficients are summarized in the table below:

— Dimension (n) — Homology Group $H_n(T^2;\mathbb{Z})$ — Description — ——————— — 0 — $\mathbb{Z}$ — The torus is path-connected. — — 1 — $\mathbb{Z}\oplus\mathbb{Z}$ — Two independent non-contractible loops (meridian and longitude). — — 2 — $\mathbb{Z}$ — The torus encloses a single 2-dimensional "void" (it is an oriented surface). — — ≥ 3 — 0 — No higher-dimensional features exist. —

Final Answer: Describing the homology of a torus Defining the CW complex structure To analyze the torus T^2, we can view it as a square with opposite sides identified. This construction allows us to define a CW complex structure, which simplifies the calculation of homology groups. The structure consists of:

- - **One 0-cell (vertex):** A single point v where all corners of the square meet.
- - **Two 1-cells (edges):** Two loops, a and b, that start and end at v.
- - **One 2-cell (face):** The interior of the square, f, attached to the edges along the boundary path $aba^{-1}b^{-1}$.

Calculating the 0th homology group H_0 The 0th homology group $H_0(T^2;\mathbb{Z})$ measures the path-connectedness of the space. Because the torus is a single, continuous piece, it has only one connected component. In algebraic terms, this corresponds to a free abelian group of rank 1.

$$H_0(T^2) \cong \mathbb{Z}$$

Calculating the 1st homology group H_1 The 1st homology group $H_1(T^2;\mathbb{Z})$ identifies 1-dimensional "holes" or loops that cannot be contracted to a point. In our CW structure, the boundary of the 2-cell f is calculated as:

$$\begin{aligned}\partial f &= a + b - a - b \\ &= 0\end{aligned}$$

Since the boundary is zero, the loops a (the meridian) and b (the longitude) are cycles that do not bound any surface. They are independent, leading to a rank-2 free abelian group:

$$H_1(T^2) \cong \mathbb{Z}\oplus\mathbb{Z}$$

Calculating the 2nd homology group H_2 The 2nd homology group $H_2(T^2;\mathbb{Z})$ represents the 2-dimensional "void" enclosed by the surface. Since the torus is a compact, oriented 2-manifold without a boundary, its top-dimensional homology group is isomorphic to $\mathbb{Z}$. In our model, the single 2-cell f acts as a cycle because its boundary is zero, and there are no 3-cells to "fill" it.

$$H_2(T^2) \cong \mathbb{Z}$$

Higher-dimensional homology groups Because the torus is a 2-dimensional manifold, it contains no cells or structures of dimension 3 or higher. Consequently, all homology groups for $n \geq 3$ are trivial.

$$H_n(T^2) = 0 \text{ for } n \geq 3$$

Answer: ¡strong¿The homology groups of the torus T^2 are $H_0(T^2) \cong \mathbb{Z}$, $H_1(T^2) \cong \mathbb{Z}^2$, $H_2(T^2) \cong \mathbb{Z}$, and $H_n(T^2) = 0$ for $n \geq 3$.

Describe relative homology.

Relative homology measures the homology of a space ***relative to a subspace***—intuitively, it captures the holes of X that are "not already present in A".

Let's go step by step.

—

1. **Setup**
   Let $A \subset X$ be a subspace of a topological space X.
   Relative homology assigns groups

$$H_n(X, A)$$

   called the **relative homology groups** of the pair (X, A).
   They measure how X differs from A topologically.
   —

2. **Definition using chains**
   In singular homology, we start with the chain groups:

$$C_n(X)$$

   generated by singular (n)-simplices in (X).
   Similarly we have

$$C_n(A)$$

   generated by simplices lying in A.
   Since $A \subset X$, we have

$$C_n(A) \subset C_n(X).$$

   The **relative chain group** is the quotient

$$C_n(X, A) = C_n(X)/C_n(A).$$

   So chains in A are treated as **zero**.
   The boundary operator descends to this quotient, giving a chain complex

$$\cdots \to C_{n+1}(X, A) \to C_n(X, A) \to C_{n-1}(X, A) \to \dots$$

   Then

$$H_n(X, A) = \frac{\ker(\partial_n : C_n(X, A) \to C_{n-1}(X, A))}{\operatorname{im}(\partial_{n+1})}.$$

   —

3. **Geometric interpretation**
   Relative cycles are chains in X whose boundary lies in A.
   So:
   * cycles may **end on** A * boundaries are those filled inside A
   Thus relative homology detects features of A that **become trivial once** A **is allowed as a boundary**.
   —

4. **Example**
   Consider the disk and its boundary:

$$X = D^2, \qquad A = S^1.$$

   Here S^1 is the boundary of the disk.
   Key result:

$$H_2(D^2, S^1) \cong \mathbb{Z}.$$

   Interpretation:
   * The disk itself represents a **relative 2-cycle** * Its boundary lies in S^1
   So it is allowed in relative homology.
   Note that:

$$H_2(D^2) = 0$$

   but relative homology detects the 2-dimensional region.
   The boundary circle is the familiar space Circle.
   —

5. **Long exact sequence of a pair**
   Relative homology fits into a fundamental structure:
   *** Long exact sequence

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X, A) \to H_{n-1}(A) \to H_{n-1}(X) \to \cdots$$

   This sequence shows how the topology of A, X, and the pair (X, A) interact.
   It is one of the most important tools in algebraic topology.
   —

6. **Intuition**
   Relative homology answers the question:
   What new holes appear when we enlarge A to X?
   or equivalently
   Which cycles in X become trivial once A is allowed as a boundary?
   —

7. **Special cases**
   If $A = \emptyset$

$$H_n(X, \emptyset) = H_n(X).$$

   So ordinary homology is a special case.
   —
   If $A = X$

$$H_n(X, X) = 0.$$

   Because everything becomes trivial.
   —

8. **Important applications**
   Relative homology is used in:

excision theorem
Mayer–Vietoris sequence
Poincaré–Lefschetz duality
manifolds with boundary
CW complex computations

For example, manifolds with boundary use

$$H_n(M, \partial M).$$

—

Key intuition

Relative homology studies **chains in X whose boundaries lie in A**.

So it measures the topology of X **relative to the part A that we treat as negligible**.

—

If you'd like, I can also explain:

• the **disk/sphere example in detail** • the **exact meaning of relative cycles** • the **excision theorem (which makes relative homology powerful)** • how relative homology appears in **degree theory and manifolds**.

Describe the Excision Theorem

The **excision theorem** is one of the central results in relative homology. It tells us that ***removing a suitable subset from both X and A does not change the relative homology***. This makes relative homology extremely powerful because it lets us simplify spaces without affecting the answer.

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1. **Statement of the excision theorem**
   Let $A \subset X$, and suppose $U \subset A$ is a subset whose closure lies in the interior of A:
   $$\overline{U} \subset \operatorname{int}(A).$$
   Then the inclusion
   $$(X - U, A - U) \hookrightarrow (X, A)$$
   induces isomorphisms
   $$H_n(X - U, A - U) \cong H_n(X, A)$$
   for all n.
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2. **Meaning of the theorem**
   We can **cut out** a region U that lies completely inside A without affecting the relative homology.
   So relative homology depends only on the portion of X **near the boundary between X and A**.
   Everything deep inside A can be removed.
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3. **Intuition**
   Relative homology measures chains in X whose boundaries lie in A.
   If a region U sits entirely inside A:
   - any chains inside U are already considered trivial
   - removing them does not affect the homology
     Thus the topology inside A is irrelevant.
     Only the **interaction between X and A** matters.

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4. **Why this theorem is useful**
   Excision allows us to **shrink complicated regions** to simpler ones.
   Many important computations in algebraic topology depend on it, including:
   - homology of spheres
   - CW complex homology
   - Mayer–Vietoris sequence
   - manifolds with boundary

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5. **Classic example: computing sphere homology**
   Consider the pair (D^n, S^{n-1}) where D^n is a disk and S^{n-1} its boundary.
   Using excision, one shows

$$H_k(D^n, S^{n-1}) = \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

This calculation helps prove the homology of spheres like the familiar boundary of a disk, the Circle.

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6. **Geometric picture**

   Imagine:

   - A is a thick region inside X
   - U is a smaller piece deep inside A

   Excision says:

   we can carve out U completely, and the relative homology stays the same.

   So the interior of A becomes irrelevant.

   Only the **interface between X and A** carries homological information.

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7. **Why excision is surprising**

   Ordinary homology **does not allow this**.

   If you remove a subset from a space, its homology usually changes.

   But relative homology is built so that:

   1. chains in A are ignored
   2. therefore parts deep inside A do not matter

   This special feature is exactly what excision formalizes.

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8. **Key idea behind the proof** *intuition*

   The proof relies on **subdividing singular simplices**.

   After enough subdivision:

   * every simplex lies either inside $X - U$ or inside A

   So chains can be replaced by equivalent chains that avoid U.

   Thus removing U does not change the homology.

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9. **Key takeaway**

   The excision theorem says:

   If a subset lies entirely inside A, we can remove it from both X and A without changing the relative homology.

   This allows complicated spaces to be **simplified locally**, which is why excision is one of the most powerful tools in algebraic topology.

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If you'd like, I can also show:

- **How excision leads to the computation of sphere homology**
- **Why $H_n(D^n, S^{n-1}) = \mathbb{Z}$**
- **How excision works inside the proof of Mayer–Vietoris**
- **A geometric picture of excision using chains and simplices**

These are usually the next big steps in mastering relative homology.